

Multidimensional Diffusive Markets

Computational Methods in Finance, Code 21040

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A 2 dimensional example I

Lognormal 2-dimensional model:

$$B(t) = e^{rt} \quad \text{riskless asset}$$

$$\frac{dS_1(t)}{S_1(t)} = \mu_1 dt + \sqrt{v_1} dW_1^{\mathbb{P}}(t) \quad \text{risky security } S_1$$

$$\frac{dS_2(t)}{S_2(t)} = \mu_2 dt + \rho\sqrt{v_2} dW_1^{\mathbb{P}}(t) + \sqrt{v_2(1-\rho^2)} dW_2^{\mathbb{P}}(t) \quad \text{risky security } S_2$$

With $W_1^{\mathbb{P}}, W_2^{\mathbb{P}}$ independent $\mathbb{P}$ -Brownian Motions. We have that

$$\text{Var} \left(\frac{dS_i(t)}{S_i(t)} \right) = v_i dt \text{ for } i = 1, 2 \text{ and } \text{Corr} \left(\frac{dS_1(t)}{S_1(t)}, \frac{dS_2(t)}{S_2(t)} \right) = \rho$$

A 2 dimensional example II

Set

$$\sigma_{11} = \sqrt{v_1} \text{ and } \sigma_{12} = 0$$

$$\sigma_{21} = \rho\sqrt{v_2} \text{ and } \sigma_{22} = \sqrt{v_2(1-\rho^2)}$$

Then

$$\frac{dS_i(t)}{S_i(t)} = \mu_i dt + \sigma_{i1} dW_1^{\mathbb{P}}(t) + \sigma_{i2} dW_2^{\mathbb{P}}(t) \text{ for } i = 1, 2$$

or

$$\frac{dS_i(t)}{S_i(t)} = \mu_i dt + \sigma_i \cdot dW^{\mathbb{P}}(t) \text{ for } i = 1, 2$$

where

$$\sigma_i = [\sigma_{i,1}, \sigma_{i,2}] \text{ for } i = 1, 2$$

$$W^{\mathbb{P}} = \begin{bmatrix} W_1^{\mathbb{P}} \\ W_2^{\mathbb{P}} \end{bmatrix} \text{ 2-dimensional B.M.}$$

A basket of lognormal risky assets I

On $(\Omega, \mathcal{F}, \mathbb{P})$: $B(t) = e^{\int_0^t r(s)ds}$ the riskless security, and $S_1, \dots, S_d$ the d risky securities ($d \geq 1$).

$$\frac{dS_i(t)}{S_i(t)} = \mu_i dt + \sigma_i \cdot dW^{\mathbb{P}}(t) \quad (1)$$

where

- $S(t) = (S_1(t), \dots, S_d(t))$
- $W^{\mathbb{P}}$ is a k -dimensional standard Brownian motion:

$$W^{\mathbb{P}} = \begin{bmatrix} W_1^{\mathbb{P}} \\ \vdots \\ W_k^{\mathbb{P}} \end{bmatrix}$$

where $W_1^{\mathbb{P}}, \dots, W_k^{\mathbb{P}}$ are *independent* (real valued) standard Brownian motions under $\mathbb{P}$.

A basket of lognormal risky assets II

- $\mu_i : \mathbb{R}^d \times \mathbb{R} \rightarrow \mathbb{R}$ are deterministic functions of the current prices $S = (S_1, \dots, S_d)^T$ and of time

In the simplest case, $\mu_i(S, t) = \mu_i$ is constant for every $i = 1, \dots, d$.

- $\sigma_i^T : \mathbb{R}^d \times \mathbb{R} \rightarrow \mathbb{R}^k$ are deterministic functions of the current prices and of time with values in $\mathbb{R}^k$.

In the simplest case, $\sigma_i^T(S, t) = [\sigma_{i,1}, \dots, \sigma_{i,k}]$ is a k -vector of constants.

$\sigma_{i,j}$ is the sensitivity of S_i 's instantaneous return to $W_j^{\mathbb{P}}$, the risk factor j

- Covariance between the instantaneous return of S_i and the one of S_j

$$\Sigma_{i,j} = \sigma_i^T \sigma_j = \sigma_i \cdot \sigma_j = \sum_{l=1}^k \sigma_{il} \sigma_{jl}$$

The No-Arbitrage analysis I

Consider the discounted gain process $\tilde{G}_i$ from holding 1 unit of S_i :

$$d\tilde{G}_i(t) = d\tilde{S}_i(t) + \tilde{S}_i(t)q_i dt$$

The RN probability $\mathbb{Q}$ is an equivalent probability under which *all* the discounted gains from risky securities $\tilde{G}_i$ are martingales.

Representation theorem of Martingales:

A process M is an $\mathbb{F}$ –martingale under a given probability $\mathbb{Q}$ if and only if it can be written as

$$dM = (\dots) \cdot dW^{\mathbb{Q}}$$

where $W^{\mathbb{Q}}$ is a $\mathbb{Q}$ -Brownian motion.

The No-Arbitrage analysis II

For each $i = 1, \dots, d$

$$\frac{d\tilde{S}_i(t)}{\tilde{S}_i(t)} = (\mu_i - r) dt + \sigma_i \cdot dW^{\mathbb{P}}(t) \quad (2)$$

$$\frac{d\tilde{G}_i(t)}{\tilde{S}_i(t)} = (\mu_i - r) dt + \sigma_i \cdot dW^{\mathbb{P}}(t) + q_i dt$$

(Multi-dimensional) Girsanov Theorem: Consider a process $\nu(t)$.
(Under suitable integrability conditions on $\nu(t)$) there exists $\mathbb{Q}$ equivalent to $\mathbb{P}$ under which

$$dW^{\mathbb{Q}}(t) = dW^{\mathbb{P}}(t) + \nu(t)dt$$

is a standard Brownian motion (see LN).

The No-Arbitrage analysis III

If there exists a $\nu \in \mathbb{R}^k$ such that for all $i = 1, \dots, d$ we have

$$\mu_i + q_i - r = \sigma_i \cdot \nu \quad (3)$$

In this case, in fact, equation (2) can be written as

$$\begin{aligned} \frac{d\tilde{G}_i(t)}{\tilde{S}_i(t)} &= (\mu_i - r) dt + \sigma_i \cdot dW^{\mathbb{P}}(t) + q_i dt \\ &= \sigma_i \cdot \left(\nu dt + dW^{\mathbb{P}}(t) \right) \end{aligned}$$

But we can now apply Girsanov theorem to find $\mathbb{Q}$ such that

$$\frac{d\tilde{G}_i(t)}{\tilde{S}_i(t)} = \sigma_i \cdot dW^{\mathbb{Q}}(t)$$

The No-Arbitrage analysis IV

the processes $\tilde{G}_i$ are martingales for all $i = 1, \dots, d!$

$\mathbb{Q}$ is RN and ν is the market price of risk (process)

Risk factors and risky assets I

If all coefficients are *constant*, but $k, d \neq 1$, then ν is a vector of $\mathbb{R}^k$ that satisfies (3)

$$\mu_i + q_i - r = \sigma_i \cdot \nu \text{ for } i = 1, \dots, d$$

If we denote the volatility matrix with

$$[\sigma_{i,j}] = \begin{bmatrix} \sigma_{1,1} & \dots & \sigma_{1,k} \\ \dots & \dots & \dots \\ \dots & \dots & \dots \\ \sigma_{d,1} & \dots & \sigma_{d,k} \end{bmatrix}$$

Equivalently

$$[\sigma_{i,j}] \begin{bmatrix} \nu_1 \\ \dots \\ \nu_k \end{bmatrix} = \begin{bmatrix} \mu_1 + q_1 - r \\ \dots \\ \mu_d + q_d - r \end{bmatrix} \quad (4)$$

and assume that $[\sigma_{i,j}]$ has full rank (securities are not redundant) we have the following rule of thumb:

Risk factors and risky assets II

- ① If $rk[\sigma_{i,j}] = d < k$ then (4) admits infinitely many solutions ν and No Arbitrage holds, but the market is incomplete (2^{nd} FTAP).
- ② If $rk[\sigma_{i,j}] = k = d$ then the market is complete and No Arbitrage holds, since (4) admits a unique solution ν (2^{nd} FTAP).

Hence, the market satisfies No Arbitrage and completeness *if and only if the number of risk factors k = the number of risky (non redundant) securities d .*

Solving the multivariate SDE I

Suppose that X satisfies

$$\begin{cases} dX_i(s) = a_i(s, X(s)) ds + b_i(s, X(s)) \cdot dW(s) \\ X(0) = X_0 \end{cases}$$

where W^P is a k -dimensional standard Brownian motion and $a_i : [0; T] \times \mathbb{R}^d \rightarrow \mathbb{R}$ and $b_i : [0; T] \times \mathbb{R}^d \rightarrow \mathbb{R}^k$ are such that X is well defined. Let $f : [0; T] \times \mathbb{R}^d \rightarrow \mathbb{R}$ be continuously differentiable, once with respect to the first variable, denoted by t , twice with respect to the second vector variable, denoted by $x \in \mathbb{R}^d$. Let

$$Y(t) = f(t; X(t))$$

Then $Y(t)$ is itself an Ito process is given by:

Solving the multivariate SDE II

$$\begin{aligned} dY(t) &= \frac{\partial f(t; X(t))}{\partial t} dt + \sum_{i=1}^d \frac{\partial f(t; X(t))}{\partial x_i} dX_i(t) + .. \\ &.. + \frac{1}{2} \sum_{i,j=1}^d \frac{\partial^2 f(t; X(t))}{\partial x_i \partial x_j} b_i(t, X(t)) \cdot b_j(t, X(t)) dt \end{aligned}$$

Applying Ito's lemma to $Y(t) = \ln(S_i(t))$, the solution of Equation (1) is

$$S_i(t) = S_i(0) \cdot e^{(\mu_i - \frac{1}{2}\sigma_i \cdot \sigma_i)t + \sigma_i \cdot W^{\mathbb{P}}(t)} \quad (5)$$

The distribution of the risky securities under the historical and the risk neutral measure I

To find the distribution under the (or an) EMM $\mathbb{Q}$ we recall that $\mu_i + q_i - r = \sigma_i^T \nu$ and that

$$W^{\mathbb{Q}}(t) = W^{\mathbb{P}}(t) + \nu t$$

is a standard Brownian motion under $\mathbb{Q}$ and

Hence

$$\left(\mu_i - \frac{1}{2} \sigma_i \cdot \sigma_i \right) t + \sigma_i \cdot W^{\mathbb{P}}(t) = \left(r - q_i - \frac{1}{2} \sigma_i \cdot \sigma_i \right) t + \sigma_i \cdot W^{\mathbb{Q}}(t)$$

so that

$$S_i(t) = S_i(0) \cdot e^{(r - q_i - \frac{1}{2} \sigma_i \cdot \sigma_i)t + \sigma_i \cdot W^{\mathbb{Q}}(t)}$$

The distribution of the risky securities under the historical and the risk neutral measure II

Since

$$\|\sigma_i\| = \sqrt{\sigma_i \cdot \sigma_i}$$

$$\sigma_i W^{\mathbb{Q}}(t) \stackrel{\mathbb{Q}}{\sim} \|\sigma_i\| B^{\mathbb{Q}}(t) \quad \text{with } B^{\mathbb{Q}} \text{ 1-dimensional } \mathbb{Q} - \text{B.M.}$$

then

$$S_i(t) \stackrel{\mathbb{Q}}{\sim} S_i(0) \cdot e^{\left(r - q_i - \frac{1}{2}\|\sigma_i\|^2\right)t + \|\sigma_i\| B^{\mathbb{Q}}(t)} \quad (6)$$

S_i is (marginally) *lognormal* with (1-dim.) *volatility* $\|\sigma_i\|$ and $\mathbb{Q}$ -drift r ($\mathbb{P}$ -drift is μ_i).

Warning: Equation (6) does not preserve the proper covariance structure with other stocks.

No Arbitrage Valuation of derivatives on several underlying assets I

We are ready to price any European derivative with payoff

$$X(T) = f(S_1(T), \dots, S_d(T))$$

at maturity T by computing

$$V(t) = \mathbb{E}_t^{\mathbb{Q}} \left[e^{-rT} X(T) \right] \text{ for any } t \in [0, T]$$

No Arbitrage Valuation of derivatives on several underlying assets II

Suppose X is an European derivative whose payoff depends at maturity only at the underlying stock prices at T :

$$X(T) = f(S_1(T), \dots, S_d(T))$$

It can be shown that

$$V(t) = \mathbb{E}^{\mathbb{Q}} \left[e^{-r(T-t)} X(T) \middle| \mathcal{F}_t \right] = F(t, S_1(t), \dots, S_d(t))$$

where F is a deterministic function of time t and current stock prices $S(t) = S_1(t), \dots, S_d(t)$.

To determine F we can use the Black-Scholes PDE.

The Black-Scholes multidimensional PDE with dividends

If security S_i provides a continuous dividend yield q_i the Black-Scholes PDE becomes

$$\frac{\partial F(t; S)}{\partial t} + \sum_{i=1}^d \frac{\partial F(t; S)}{\partial S_i} \cdot S_i (r - q_i) + \frac{1}{2} \sum_{i,j=1}^d \frac{\partial^2 F(t; S)}{\partial S_i \partial S_j} \sigma_i \cdot \sigma_j S_i S_j = F(t, S)r$$

for any $S = (S_1, \dots, S_d) \in \mathbb{R}_+^d$ and for any $t \in (0, T)$ with the boundary condition

$$F(T, S) = f(S) \text{ for any } S = (S_1, \dots, S_d) \in \mathbb{R}_+^d$$

Replicating strategy

$$h_i = \frac{\partial F}{\partial S_i} \text{ for } i = 1, \dots, d \text{ hedge ratios of } F$$

The self-financing strategy that replicates F is given by

$$\begin{aligned}\vartheta_0(t) &= \left(F(t, S(t)) - \underbrace{\sum_{i=1}^d \frac{\partial F}{\partial S_i}(t, S(t)) S_i(t)}_{h_i(t)} \right) \frac{1}{B(t)} \\ \vartheta_i(t) &= \underbrace{\frac{\partial F}{\partial S_i}(t, S(t))}_{h_i(t)} \text{ for } i = 1, \dots, d\end{aligned}$$